

```
import numpy as np
import pandas as pd
```

▼ Creating DataFrame

```
# using lists
student_data = [
    [100,80,10],
    [90,70,7],
    [120,100,14],
    [80,50,2]
]

pd.DataFrame(student_data,columns=['iq','marks','package'])
```

	iq	marks	package
0	100	80	10
1	90	70	7
2	120	100	14
3	80	50	2

```
# using dicts

student_dict = {
    'name':['nitish','ankit','rupesh','rishabh','amit','ankita'],
    'iq':[100,90,120,80,0,0],
    'marks':[80,70,100,50,0,0],
    'package':[10,7,14,2,0,0]
}

students = pd.DataFrame(student_dict)
students.set_index('name',inplace=True)
students
```

	iq	marks	package
name			
nitish	100	80	10
ankit	90	70	7
rupesh	120	100	14
rishabh	80	50	2
amit	0	0	0
ankita	0	0	0

```
# using read_csv
movies = pd.read_csv('movies.csv')
movies
```

	title_x	imdb_id	poster_path		wiki_link	title_y	original_title
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike	Uri: The Surgical Strike	Uri: The Surgical Strike
1	Battalion 609	tt9472208		NaN	https://en.wikipedia.org/wiki/Battalion_609	Battalion 609	Battalion 609
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Accidental_P...	The Accidental Prime Minister	The Accidental Prime Minister	The Accidental Prime Minister
3	Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Why_Cheat_India	Why Cheat India	Why Cheat India	Why Cheat India
4	Evening Shadows	tt6028796		NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows
...	...	...	...	...	...	...	...
1624	Tera Mera Saath Rahen	tt0301250	https://upload.wikimedia.org/wikipedia/en/2/2b...	https://en.wikipedia.org/wiki/Tera_Mera_Saath_...	Tera Mera Saath Rahen	Tera Mera Saath Rahen	Tera Mera Saath Rahen
1625	Yeh Zindagi Ka Safar	tt0298607	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Yeh_Zindagi_Ka_S...	Yeh Zindagi Ka Safar	Yeh Zindagi Ka Safar	Yeh Zindagi Ka Safar
1626	Sabse Bada Sukh	tt0069204		NaN	https://en.wikipedia.org/wiki/Sabse_Bada_Sukh	Sabse Bada Sukh	Sabse Bada Sukh
1627	Daaka	tt10833860	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Daaka	Daaka	Daaka	Daaka
1628	Humsafar	tt2403201	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Humsafar	Humsafar	Humsafar	Humsafar

1629 rows × 8 columns

```
ipl = pd.read_csv('ipl-matches.csv')
ipl
```

	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	SuperOver	W
0	1312200	Ahmedabad	2022-05-29	2022	Final	Rajasthan Royals	Gujarat Titans	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	bat	N	
1	1312199	Ahmedabad	2022-05-27	2022	Qualifier 2	Royal Challengers Bangalore	Rajasthan Royals	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	field	N	
2	1312198	Kolkata	2022-05-25	2022	Eliminator	Royal Challengers Bangalore	Lucknow Super Giants	Eden Gardens, Kolkata	Lucknow Super Giants	field	N	
3	1312197	Kolkata	2022-05-24	2022	Qualifier 1	Rajasthan Royals	Gujarat Titans	Eden Gardens, Kolkata	Gujarat Titans	field	N	
4	1304116	Mumbai	2022-05-22	2022	70	Sunrisers Hyderabad	Punjab Kings	Wankhede Stadium, Mumbai	Sunrisers Hyderabad	bat	N	F
...	...	...	...	...	...	...	...	...	...	...	...	...
945	335986	Kolkata	2008-04-20	2007/08	4	Kolkata Knight Riders	Deccan Chargers	Eden Gardens	Deccan Chargers	bat	N	K
946	335985	Mumbai	2008-04-20	2007/08	5	Mumbai Indians	Royal Challengers Bangalore	Wankhede Stadium	Mumbai Indians	bat	N	
947	335984	Delhi	2008-04-19	2007/08	3	Delhi Daredevils	Rajasthan Royals	Feroz Shah Kotla	Rajasthan Royals	bat	N	
948	335983	Chandigarh	2008-04-19	2007/08	2	Kings XI Punjab	Chennai Super Kings	Punjab Cricket Association Stadium, Mohali	Chennai Super Kings	bat	N	
949	335982	Bangalore	2008-04-18	2007/08	1	Royal Challengers Bangalore	Kolkata Knight Riders	M Chinnaswamy Stadium	Royal Challengers Bangalore	field	N	K

950 rows × 20 columns

▼ DataFrame Attributes and Methods

```
# shape
movies.shape
ipl.shape
```

(950, 20)

```
# dtypes
movies.dtypes
ipl.dtypes
```

ID	int64
City	object
Date	object
Season	object
MatchNumber	object
Team1	object

```

Team2          object
Venue          object
TossWinner     object
TossDecision   object
SuperOver      object
WinningTeam    object
WonBy          object
Margin         float64
method         object
Player_of_Match object
Team1Players   object
Team2Players   object
Umpire1        object
Umpire2        object
dtype: object

```

```

# index
movies.index
ipl.index

```

```
RangeIndex(start=0, stop=950, step=1)
```

```

# columns
movies.columns
ipl.columns
student.columns

```

```
Index(['iq', 'marks', 'package'], dtype='object')
```

```

# values
student.values
ipl.values

```

```

array([[1312200, 'Ahmedabad', '2022-05-29', ...,
        ["WP Saha", 'Shubman Gill', 'MS Wade', 'HH Pandya', 'DA Miller', 'R Tewatia', 'Rashid Khan', 'R Sai Kishore', 'LH
Ferguson', 'Yash Dayal', 'Mohammed Shami']],
        ['CB Gaffaney', 'Nitin Menon'],
        [1312199, 'Ahmedabad', '2022-05-27', ...,
        ["YBK Jaiswal", 'JC Buttler', 'SV Samson', 'D Padikkal', 'SO Hetmyer', 'R Parag', 'R Ashwin', 'TA Boult', 'YS Chahal',
'M Prasad Krishna', 'OC McCoy']],
        ['CB Gaffaney', 'Nitin Menon'],
        [1312198, 'Kolkata', '2022-05-25', ...,
        ["Q de Kock", 'KL Rahul', 'M Vohra', 'DJ Hooda', 'MP Stoinis', 'E Lewis', 'KH Pandya', 'PVD Chameera', 'Mohsin Khan',
'Avesh Khan', 'Ravi Bishnoi']],
        ['J Madanagopal', 'MA Gough'],
        ...,
        [335984, 'Delhi', '2008-04-19', ...,
        ["T Kohli", 'YK Pathan', 'SR Watson', 'M Kaif', 'DS Lehmann', 'RA Jadeja', 'M Rawat', 'D Salunkhe', 'SK Warne', 'SK
Trivedi', 'MM Patel']],
        ['Aleem Dar', 'GA Pratapkumar'],
        [335983, 'Chandigarh', '2008-04-19', ...,
        ["PA Patel", 'ML Hayden', 'MEK Hussey', 'MS Dhoni', 'SK Raina', 'JDP Oram', 'S Badrinath', 'Joginder Sharma', 'P
Amarnath', 'MS Gony', 'M Muralitharan']],
        ['MR Benson', 'SL Shastri'],
        [335982, 'Bangalore', '2008-04-18', ...,
        ["SC Ganguly", 'BB McCullum', 'RT Ponting', 'DJ Hussey', 'Mohammad Hafeez', 'LR Shukla', 'WP Saha', 'AB Agarkar', 'AB
Dinda', 'M Kartik', 'I Sharma']],
        ['Asad Rauf', 'RE Koertzen']], dtype=object)

```

```

# head and tail
movies.head(2)

```

	title_x	imdb_id	poster_path		wiki_link	title_y	original_title
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike	Uri: The Surgical Strike	Uri: The Surgical Strike
1	Battalion 609	tt9472208		NaN	https://en.wikipedia.org/wiki/Battalion_609	Battalion 609	Battalion 609

ipl.tail(2)

	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	SuperOver	Winner
948	335983	Chandigarh	2008-04-19	2007/08	2	Kings XI Punjab	Chennai Super Kings	Punjab Cricket Association Stadium, Mohali	Chennai Super Kings	bat	N	Chennai Super Kings
949	335982	Bangalore	2008-04-18	2007/08	1	Royal Challengers Bangalore	Kolkata Knight Riders	M Chinnaswamy Stadium	Royal Challengers Bangalore	field	N	Kolkata Knight Riders

sample
ipl.sample(5)

	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	SuperOver	Winner
336	1082628	Mumbai	2017-05-01	2017	38	Mumbai Indians	Royal Challengers Bangalore	Wankhede Stadium	Royal Challengers Bangalore	bat	N	Royal Challengers Bangalore
98	1254107	Sharjah	2021-09-25	2021	37	Punjab Kings	Sunrisers Hyderabad	Sharjah Cricket Stadium	Sunrisers Hyderabad	field	N	Punjab Kings
890	392182	Cape Town	2009-04-18	2009	2	Royal Challengers Bangalore	Rajasthan Royals	Newlands	Royal Challengers Bangalore	bat	N	Chennai Super Kings
157	1216533	Abu Dhabi	2020-10-19	2020/21	37	Chennai Super Kings	Rajasthan Royals	Sheikh Zayed Stadium	Chennai Super Kings	bat	N	Chennai Super Kings
386	980991	Chandigarh	2016-05-15	2016	46	Kings XI Punjab	Sunrisers Hyderabad	Punjab Cricket Association IS Bindra Stadium, ...	Kings XI Punjab	bat	N	Kings XI Punjab

```
# info
movies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1629 entries, 0 to 1628
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   title_x               1629 non-null   object
1   imdb_id               1629 non-null   object
2   poster_path           1526 non-null   object
3   wiki_link             1629 non-null   object
4   title_y               1629 non-null   object
5   original_title        1629 non-null   object
6   is_adult              1629 non-null   int64
7   year_of_release       1629 non-null   int64
8   runtime               1629 non-null   object
9   genres                1629 non-null   object
10  imdb_rating            1629 non-null   float64
11  imdb_votes             1629 non-null   int64
12  story                  1609 non-null   object
13  summary                1629 non-null   object
14  tagline                557 non-null    object
15  actors                 1624 non-null   object
16  wins_nominations       707 non-null    object
17  release_date           1522 non-null   object
dtypes: float64(1), int64(3), object(14)
memory usage: 229.2+ KB
```

```
ipl.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 950 entries, 0 to 949
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                    950 non-null    int64
1   City                  899 non-null    object
2   Date                  950 non-null    object
3   Season                950 non-null    object
4   MatchNumber           950 non-null    object
5   Team1                 950 non-null    object
6   Team2                 950 non-null    object
7   Venue                 950 non-null    object
8   TossWinner            950 non-null    object
9   TossDecision           950 non-null    object
10  SuperOver              946 non-null    object
11  WinningTeam            946 non-null    object
12  WonBy                  950 non-null    object
13  Margin                 932 non-null    float64
14  method                 19 non-null     object
15  Player_of_Match        946 non-null    object
16  Team1Players           950 non-null    object
17  Team2Players           950 non-null    object
18  Umpire1                950 non-null    object
19  Umpire2                950 non-null    object
dtypes: float64(1), int64(1), object(18)
memory usage: 148.6+ KB
```

```
# describe
movies.describe()
```

	is_adult	year_of_release	imdb_rating	imdb_votes
count	1629.0	1629.000000	1629.000000	1629.000000
mean	0.0	2010.263966	5.557459	5384.263352
std	0.0	5.381542	1.567609	14552.103231
min	0.0	2001.000000	0.000000	0.000000
25%	0.0	2005.000000	4.400000	233.000000
50%	0.0	2011.000000	5.600000	1000.000000
75%	0.0	2015.000000	6.800000	4287.000000
max	0.0	2019.000000	9.400000	310481.000000

```
ipl.describe()
```

	ID	Margin
count	9.500000e+02	932.000000
mean	8.304852e+05	17.056867
std	3.375678e+05	21.633109
min	3.359820e+05	1.000000
25%	5.012612e+05	6.000000
50%	8.297380e+05	8.000000
75%	1.175372e+06	19.000000
max	1.312200e+06	146.000000

```
# isnull
movies.isnull().sum()
```

```
title_x          0
imdb_id          0
poster_path     103
wiki_link        0
title_y          0
original_title   0
is_adult         0
year_of_release  0
runtime          0
genres           0
imdb_rating      0
imdb_votes       0
story           20
summary          0
tagline         1072
actors           5
wins_nominations 922
release_date     107
dtype: int64
```

```
# duplicated
movies.duplicated().sum()
```

0

```
students.duplicated().sum()
```

1

```
# rename
students
```

	iq	percent	lpa
0	100	80	10
1	90	70	7
2	120	100	14
3	80	50	2
4	0	0	0
5	0	0	0

```
students.rename(columns={'marks':'percent','package':'lpa'},inplace=True)
```

Math Methods

```
# sum -> axis argument
students.sum(axis=0)
```

```
iq          390
percent     300
```

```
lpa          33
dtype: int64
```

```
students.mean(axis=1)
```

```
0    63.333333
1    55.666667
2    78.000000
3    44.000000
4     0.000000
5     0.000000
dtype: float64
```

```
students.var()
```

```
iq          2710.0
percent     1760.0
lpa          33.5
dtype: float64
```

Start coding or [generate](#) with AI.

▼ Selecting cols from a DataFrame

```
# single cols
movies['title_x']
```

```
0          Uri: The Surgical Strike
1          Battalion 609
2    The Accidental Prime Minister (film)
3          Why Cheat India
4          Evening Shadows
...
1624          Tera Mera Saath Rahen
1625          Yeh Zindagi Ka Safar
1626          Sabse Bada Sukh
1627          Daaka
1628          Humsafar
Name: title_x, Length: 1629, dtype: object
```

```
ipl['Venue']
```

```
0          Narendra Modi Stadium, Ahmedabad
1          Narendra Modi Stadium, Ahmedabad
2          Eden Gardens, Kolkata
3          Eden Gardens, Kolkata
4          Wankhede Stadium, Mumbai
...
945          Eden Gardens
946          Wankhede Stadium
947          Feroz Shah Kotla
948    Punjab Cricket Association Stadium, Mohali
949          M Chinnaswamy Stadium
Name: Venue, Length: 950, dtype: object
```

```
# multiple cols
movies[['year_of_release', 'actors', 'title_x']]
```


	year_of_release	actors	title_x
0	2019	Vicky Kaushal Paresh Rawal Mohit Raina Yami Ga...	Uri: The Surgical Strike
1	2019	Vicky Ahuja Shoaib Ibrahim Shrikant Kamat Elen...	Battalion 609
2	2019	Anupam Kher Akshaye Khanna Aahana Kumra Atul S...	The Accidental Prime Minister (film)
3	2019	Emraan Hashmi Shreya Dhanwanthary Snighdadeep ...	Why Cheat India
4	2018	Mona Ambegaonkar Ananth Narayan Mahadevan Deva...	Evening Shadows
...	...	...	...
1624	2001	Ajay Devgn Sonali Bendre Namrata Shirodkar Pre...	Tera Mera Saath Rahen
1625	2001	Ameesha Patel Jimmy Sheirgill Nafisa Ali Gulsh...	Yeh Zindagi Ka Safar
1626	2018	Vijay Arora Asrani Rajni Bala Kumud Damle Utpa...	Sabse Bada Sukh
1627	2019	Gippy Grewal Zareen Khan	Daaka
1628	2011	Fawad Khan	Humsafar

1629 rows × 3 columns

```
ipl[['Team1','Team2','WinningTeam']]
```

	Team1	Team2	WinningTeam
0	Rajasthan Royals	Gujarat Titans	Gujarat Titans
1	Royal Challengers Bangalore	Rajasthan Royals	Rajasthan Royals
2	Royal Challengers Bangalore	Lucknow Super Giants	Royal Challengers Bangalore
3	Rajasthan Royals	Gujarat Titans	Gujarat Titans
4	Sunrisers Hyderabad	Punjab Kings	Punjab Kings
...	...	...	...
945	Kolkata Knight Riders	Deccan Chargers	Kolkata Knight Riders
946	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore
947	Delhi Daredevils	Rajasthan Royals	Delhi Daredevils
948	Kings XI Punjab	Chennai Super Kings	Chennai Super Kings
949	Royal Challengers Bangalore	Kolkata Knight Riders	Kolkata Knight Riders

950 rows × 3 columns

✓ Selecting rows from a DataFrame

- **iloc** - searches using index positions
- **loc** - searches using index labels

```
# single row
movies.iloc[5]
```

```

title_x          Soni (film)
imdb_id          tt6078866
poster_path      https://upload.wikimedia.org/wikipedia/en/thum...
wiki_link        https://en.wikipedia.org/wiki/Soni_(film)
title_y          Soni
original_title   Soni
is_adult         0
year_of_release  2018
runtime         97
genres           Drama
imdb_rating      7.2
imdb_votes       1595
story            Soni a young policewoman in Delhi and her su...
summary          While fighting crimes against women in Delhi ...
tagline          NaN
actors           Geetika Vidya Ohlyan|Saloni Batra|Vikas Shukla...
wins_nominations 3 wins & 5 nominations
release_date     18 January 2019 (USA)
Name: 5, dtype: object

```

```
# multiple row
movies.iloc[:5]
```

	title_x	imdb_id	poster_path	wiki_link	title_y	original_title
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike	Uri: The Surgical Strike
1	Battalion 609	tt9472208	NaN	https://en.wikipedia.org/wiki/Battalion_609	Battalion 609	Battalion 609
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Accidental_P...	The Accidental Prime Minister	The Accidental Prime Minister
3	Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Why_Cheat_India	Why Cheat India	Why Cheat India
4	Evening Shadows	tt6028796	NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows

```
# fancy indexing
movies.iloc[[0,4,5]]
```

	title_x	imdb_id	poster_path	wiki_link	title_y	original_title
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike	Uri: The Surgical Strike
4	Evening Shadows	tt6028796	NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows
5	Soni (film)	tt6078866	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Soni_(film)	Soni	Soni

```
# loc
students
```

	iq	marks	package
name			
nitish	100	80	10
ankit	90	70	7
rupesh	120	100	14
rishabh	80	50	2
amit	0	0	0
ankita	0	0	0

```
students.loc['nitish']
```

```
iq      100
marks   80
package 10
Name: nitish, dtype: int64
```

```
students.loc['nitish':'rishabh':2]
```

	iq	marks	package
name			
nitish	100	80	10
rupesh	120	100	14

```
students.loc[['nitish','ankita','rupesh']]
```

	iq	marks	package
name			
nitish	100	80	10
ankita	0	0	0
rupesh	120	100	14

```
students.iloc[[0,3,4]]
```

	iq	marks	package
name			
nitish	100	80	10
rishabh	80	50	2
amit	0	0	0

✓ Selecting both rows and cols

```
movies.iloc[0:3,0:3]
```

	title_x	imdb_id	poster_path
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...
1	Battalion 609	tt9472208	NaN
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...

```
movies.loc[0:2,'title_x':'poster_path']
```

	title_x	imdb_id	poster_path
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...
1	Battalion 609	tt9472208	NaN
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...

Filtering a DataFrame

```
ipl.head(2)
```

	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	SuperOver	Winning
0	1312200	Ahmedabad	2022-05-29	2022	Final	Rajasthan Royals	Gujarat Titans	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	bat	N	Gu
1	1312199	Ahmedabad	2022-05-27	2022	Qualifier 2	Royal Challengers Bangalore	Rajasthan Royals	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	field	N	Raja

```
# find all the final winners
mask = ipl['MatchNumber'] == 'Final'
new_df = ipl[mask]
new_df[['Season', 'WinningTeam']]

ipl[ipl['MatchNumber'] == 'Final'][['Season', 'WinningTeam']]
```

	Season	WinningTeam
0	2022	Gujarat Titans
74	2021	Chennai Super Kings
134	2020/21	Mumbai Indians
194	2019	Mumbai Indians
254	2018	Chennai Super Kings
314	2017	Mumbai Indians
373	2016	Sunrisers Hyderabad
433	2015	Mumbai Indians
492	2014	Kolkata Knight Riders
552	2013	Mumbai Indians
628	2012	Kolkata Knight Riders
702	2011	Chennai Super Kings
775	2009/10	Chennai Super Kings
835	2009	Deccan Chargers
892	2007/08	Rajasthan Royals

```
# how many super over finishes have occurred
ipl[ipl['SuperOver'] == 'Y'].shape[0]
```

14

```
# how many matches has csk won in kolkata
ipl[(ipl['City'] == 'Kolkata') & (ipl['WinningTeam'] == 'Chennai Super Kings')].shape[0]
```

5

```
# toss winner is match winner in percentage
(ipl[ipl['TossWinner'] == ipl['WinningTeam']].shape[0]/ipl.shape[0])*100
```

51.473684210526315

```
# movies with rating higher than 8 and votes>10000
movies[(movies['imdb_rating'] > 8.5) & (movies['imdb_votes'] > 10000)].shape[0]
```

0

```
# Action movies with rating higher than 7.5
# mask1 = movies['genres'].str.split('|').apply(lambda x:'Action' in x)
mask1 = movies['genres'].str.contains('Action')
mask2 = movies['imdb_rating'] > 7.5
```

movies[mask1 & mask2]

	title_x	imdb_id	poster_path	wiki_link	title_y
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike
41	Family of Thakurganj	tt8897986	https://upload.wikimedia.org/wikipedia/en/9/99...	https://en.wikipedia.org/wiki/Family_of_Thakur...	Family of Thakurganj
84	Mukkabaaz	tt7180544	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Mukkabaaz	The Brawler
106	Raazi	tt7098658	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Raazi	Raazi
110	Parmanu: The Story of Pokhran	tt6826438	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Parmanu:_The_Sto...	Parmanu: The Story of Pokhran
112	Bhavesh Joshi Superhero	tt6129302	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Bhavesh_Joshi_Su...	Bhavesh Joshi Superhero
169	The Ghazi Attack	tt6299040	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Ghazi_Attack...	The Ghazi Attack
219	Raag Desh (film)	tt6080746	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Raagdes	Raag Desh
258	Irudhi Suttru	tt5310090	https://upload.wikimedia.org/wikipedia/en/f/fe...	https://en.wikipedia.org/wiki/Saala_Khadoos	Saala Khadoos
280	Laal Rang	tt5600714	NaN	https://en.wikipedia.org/wiki/Laal_Rang	Laal Rang

```
# write a function that can return the track record of 2 teams against each other
```

Adding new cols

354	Dangal (film)	tt5074352	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Dangal_(film)	Dangal
-----	---------------	-----------	---	---	--------

```
# completely new
movies['Country'] = 'India'
movies.head()
```

	title_x	imdb_id	poster_path	wiki_link	title_y	original_title
362	Bhajaan	tt3863552	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Bajrangi_Bhajaan	Bhajaan	Bhajaan
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike	Uri: The Surgical Strike
1393	Battalion 609 Detective Byomkesh Bakshy!	tt3447364	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Battalion_609_Detective_Byomkesh_Bakshy!	Battalion 609 Detective Byomkesh Bakshy!	Detective Battalion 609 Bakshy!
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Accidental_P...	The Accidental Prime Minister	The Accidental Prime Minister
536	Why Chhota India Hai	tt1002083	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Why_Chhota_India_Hai	Why Chhota India	Why Chhota India
4	Evening Shadows	tt6028796	NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows
625	Madras Cafe	tt2855648	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Madras_Cafe	Madras Cafe	Madras Cafe

```
# from existing ones
movies.dropna(inplace=True)
```

668	Paan Singh Tomar	tt1620933	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Paan_Singh_Tomar...	Paan Singh Tomar	Paan Singh Tomar
693	Gangs of Wasseypur	tt1954470	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Gangs_of_Wasseypur	Gangs of Wasseypur	Gangs of Wasseypur
694	Gangs of Wasseypur – Part 2	tt1954470	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Gangs_of_Wasseyp...	Gangs of Wasseypur	Gangs of Wasseypur
982	Jodhaa Akbar	tt0449994	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Jodhaa_Akbar	Jodhaa Akbar	Jodhaa Akbar
1039	1971 (2007 film)	tt0983990	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/1971_(2007_film)	1971	1971
1058	Black Friday (2007 film)	tt0400234	https://upload.wikimedia.org/wikipedia/en/5/58...	https://en.wikipedia.org/wiki/Black_Friday_(20...	Black Friday	Black Friday
1188	Omkaara (2006 film)	tt0488414	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Omkara_(2006_film)	Omkaara	Omkaara
1293	Sarkar (2005 film)	tt0432047	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Sarkar_(2005_film)	Sarkar	Sarkar

	title_x	imdb_id	poster_path	wiki_link	title_y	original_title
11	Gully Boy	tt2395469	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Gully_Boy	Gully Boy	Gully Boy
1361	Lakshya (film)	tt0323013	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Lakshya_(film)		Lakshya
34	Yeh Hai India	tt5525846	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Yeh_Hai_India	Yeh Hai India	Yeh Hai India
1432	Gangaajal	tt0373856	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Gangaajal		Gangaajal
37	Article 15 (film)	tt10324144	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Article_15_(film)	Article 15	Article 15
1554	The Legend of Bhagat Singh	tt0319736	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Legend_of_Bh...		The Legend of Bhagat Singh
87	Aiyaary	tt6774212	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Aiyaary	Aiyaary	Aiyaary
1607	Nayak (2001 film)	tt0291376	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Nayak_(2001_film)		Nayak: The
96	Raid (2018 film)	tt7363076	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Raid_(2018_film)	Raid	Raid

movies.info()

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 298 entries, 11 to 1623
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   title_x               298 non-null    object
1   imdb_id               298 non-null    object
2   poster_path           298 non-null    object
3   wiki_link             298 non-null    object
4   title_y               298 non-null    object
5   original_title        298 non-null    object
6   is_adult              298 non-null    int64
7   year_of_release       298 non-null    int64
8   runtime               298 non-null    object
9   genres                298 non-null    object
10  imdb_rating           298 non-null    float64
11  imdb_votes            298 non-null    int64
12  story                 298 non-null    object
13  summary               298 non-null    object
14  tagline               298 non-null    object
15  actors                298 non-null    object
16  wins_nominations      298 non-null    object
17  release_date          298 non-null    object
18  Country               298 non-null    object
dtypes: float64(1), int64(3), object(15)
memory usage: 46.6+ KB
```

Important DataFrame Functions

```
# astype
ipl.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 950 entries, 0 to 949
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
```

```
---  ---
0   ID      950 non-null   int64
1   City    899 non-null   object
2   Date    950 non-null   object
3   Season  950 non-null   object
```